

JS → programming language → high lvl vs low level
↓
human readable binary
0&1

why JS?

Frontend, Backend, JS → Browser

where JS run

↳ Browser, html with script tag
→ ME → node.js

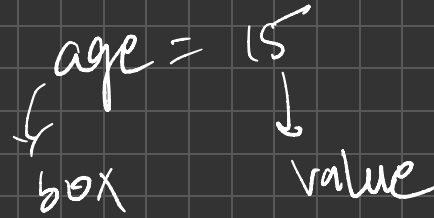
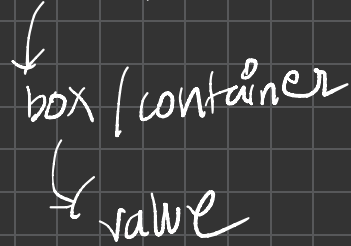
Telegram



Discord



variables in JS



3 keywords

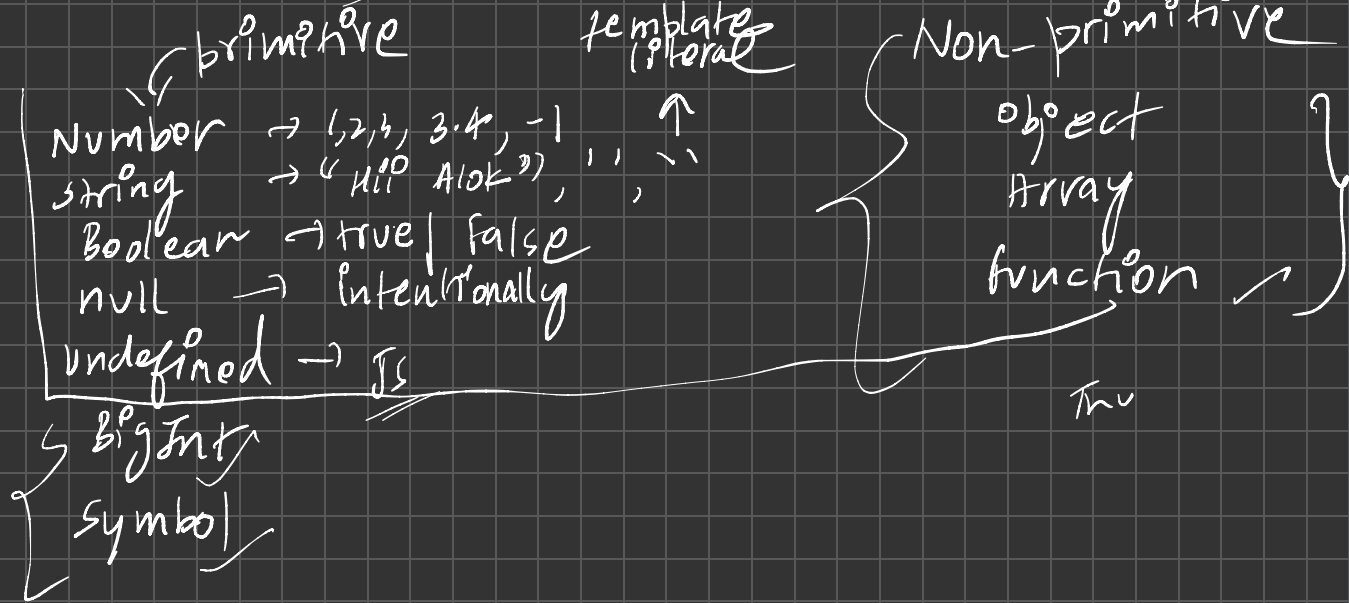
var, let & const

↳ redeclaration
update

↳ update

↳ no-redeclaration
no update

Data types



operators

Symbol $\rightarrow$ do operands ke
beech operation perform krta
hai

$\rightarrow$ Arithmetic operators
(Maths)

$+$, $-$, $*$, $/$, $\%$, $**$

$\downarrow$ addition $\downarrow$ multiply $\downarrow$ division (quotient) $\downarrow$ modulus (remainder) $\rightarrow$ exponentiation (power)

subtraction

2^3

```
let num1 = 2;
```

```
let num2 = 4;
```

```
console.log(num1 + num2);
```

```
console.log(num1 - num2);
```

```
console.log(num1 * num2);
```

```
console.log(num2 / num1);
```

```
console.log(num2 % num1);
```

```
console.log(num1 ** num2);
```

6 ✓

-2 ✓

8 ✓

2 ✓

0 ✓

16

left right operand

$2^4 \rightarrow$
 $2 \leftarrow * * 4$

$2 \overline{) 361}$
 $\underline{2}$
 $\underline{1}$ ✓

dividend
divisor
quotient
remainder

$2 \overline{) 42}$
 $\underline{4}$
 $\underline{0}$ ✓

$\div$ → divide / quotient

$\div$ → modulus
remainder

JS case sensitive

```
// case sensitive  
// teeno variable different hai  
  
let NAME = "saif"  
let name = "juhi"  
let Name = "karan"
```

(++) increment & decrement (--)
↳ post
↳ pre

+=, -

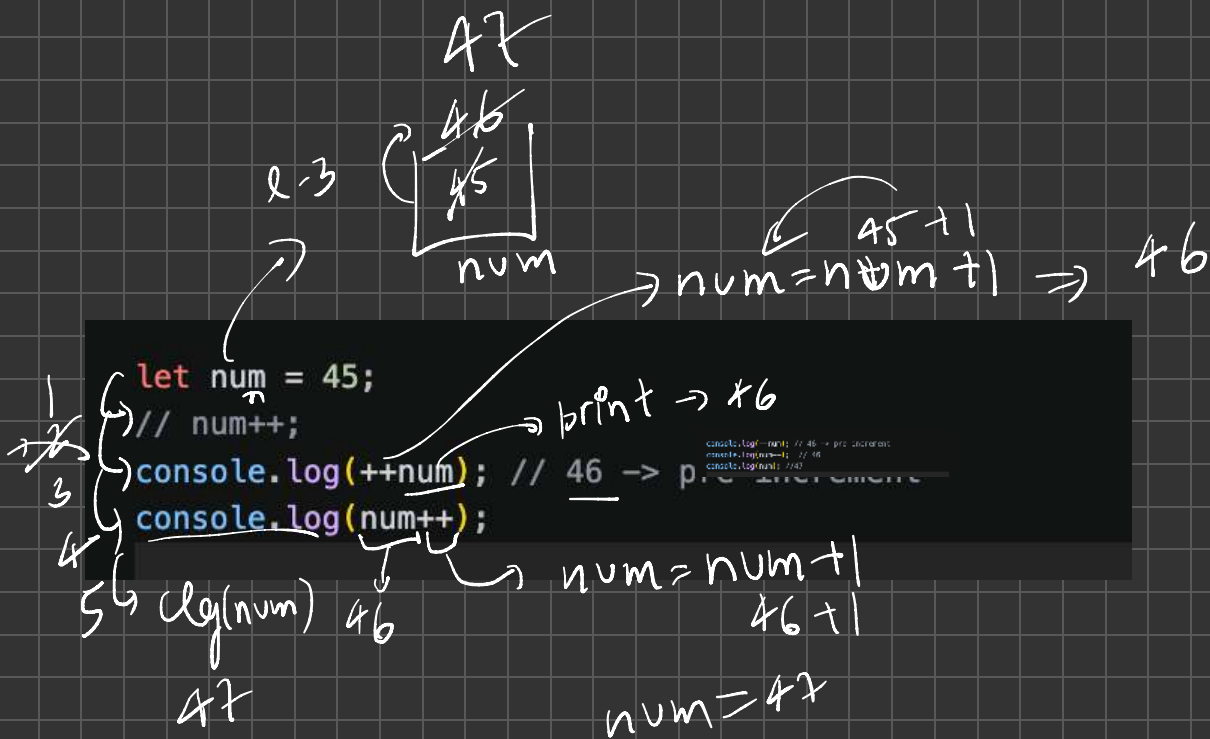
num++
↳ num = num + 1;

num++;

```
let num = 45; ✓  
num++;  
console.log(num);  
|
```

```
let num = 45; 46  
// num++;  
→ console.log(++num); // 46 → pre increment  
console.log(num++);  
                    ↓  
                    46
```

```
let num = 45; ✓  
// num++;  
// console.log(++num); // 46 → pre increment  
console.log(num++); 46  
                    ↓  
                    45  
→ log(num) → 46
```



num = ~~45~~ ~~44~~ 43

```
console.log(--num); // 44 → pre increment  
console.log(num--); // 44  
console.log(num); // 43
```

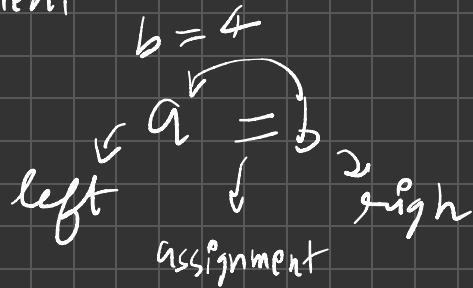
↓
43

num--
↳ num = num - 1

Assignment operator

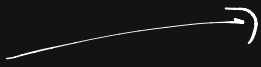
=, +=, -=, *=, /=, %=, **=

assignment



$a = 4$

```
✓ let num = 2;  
  num += 5;  
  console.log(num);
```



$num = num + 5;$

$num += 5$

$num = +5$



$num = +5$

```

let num = 2;
num += 5;    → num = num + 5
console.log(num);    → 7
num -= 5;    → num = num - 5
console.log(num);    → 2
num *= 5;    → num = num * 5
console.log(num);    → 10

```

7 2 10

```

num /= 5;    → num = num / 5
console.log(num);    → 2
num %= 5;    → num = num % 5
console.log(num);    → 2
num **= 5;   → num = num ** 5
console.log(num);    → 32

```

7 2 10 2 2 32
2 % 5 = 2

```
let num = 2;  
num += 5;  
console.log(num);  
num -= 5;  
console.log(num);  
num *= 5;  
console.log(num);  
num /= 5;  
console.log(num);  
num %=5;  
console.log(num);  
num **=5 ;  
console.log(num);
```

comparison operator $\rightarrow$ always return boolean

$>$ $\rightarrow$ greater than

$<$ $\rightarrow$ less than

$>=$ $\rightarrow$ greater than or equal

$<=$ $\rightarrow$ less than or equal

$=$ $\rightarrow$ loose equality

$===$ $\rightarrow$ strict equality

$!=$ $\rightarrow$ not equals to

```
const num1 = 3;  
const num2 = 6;
```

```
console.log(3 > 6)  
console.log(3 < 6)  
console.log(3 >= 6)  
console.log(3 <= 6)  
console.log(3 == 6)  
console.log(3 != 6)
```

false
T
F
T
F
T

F T F T F T

3 $\neq$ 5
0 $\neq$

$b > b$	F
$b < b$	F
$b == b$	T
$b >= b$	T
$b <= b$	T

Value must same

→ loose ⁽⁼⁼⁾ equality

→ strict ⁽⁼⁼⁼⁾ equality

type check nhi krta

```
console.log("5" == 5);
```

humesha type check krta hai and type must be same

```
// loose equality  
console.log("5" == 5);
```

```
// strict equality  
console.log("5" === 5);
```

true
false

Logical operators

&& → AND → dono true hone chahiye

|| → OR → ek bhi true hoga chalega

! → NOT → opposite krdo

Truth Table

<u>$\rightarrow$</u> A	B	
True	True	$\rightarrow$ True ✓
True	False	$\rightarrow$ false
False	True	$\rightarrow$ false
False	False	$\rightarrow$ false

club

$18 \geq$
Id ✓

```
const age = 18;
const hasId = true;
const canEnterClub = (age >= 18 && hasId === true);
console.log(canEnterClub);
```

online
vbi ✓
cards X

OR Truth Table

<u>A</u>	<u>B</u>	
True	True	$\rightarrow$ True
True	False	$\rightarrow$ True
False	True	$\rightarrow$ True
false	False	$\rightarrow$ false ✓

! $\rightarrow$ opposite

Truthy & falsy value

falsy

- 1 0 → zero
- 2 "" → empty string
- 3 null →
- 4 undefined
- 5 NaN
- 6 false

Truthy

"hi"

5666

true

True $\rightarrow 1$
False $\rightarrow 0$

Operator precedence
BODMAS

$$2 + 3 \times 4$$

Conditionals

↳ condition

Bank Account / SPM card

18+

If else

condition

↳ boolean

if {

} {

} else {

↳ no condition

}

youtube

↳ logged in

↳ True

↳ like, comment ✓

↳ not logged in

↳ like X, comment X

False
if (true) {



→ True block

} else {



✓ false block

}

if / else if / else

~~key~~ key

fri === mon → false

fri === Tue

①

```
let day = "fri";  
if (day === "mon") {  
  console.log("1st day of the week");  
} else if (day === "tue") {  
  console.log("2nd day of the week");  
} else if (day === "wed") {  
  console.log("3rd day of the week");  
} else if (day === "thu") {  
  console.log("4th day of the week");  
} else if (day === "fri") {  
  console.log("5th day of the week");  
} else if (day === "sat") {  
  console.log("6th day of the week");  
} else if (day === "sun") {  
  console.log("7th day of the week");  
} else {  
  console.log("Wrong day");  
}
```

90 → A
80-90 → B
70-80 → C
60-70 → D
≤ 60 → F

Nested if else

if logged in
↳ if subscribed ✓

else →

else

→ please

↳ Netflix

↳ subscription

↳ free user

code readability

Switch case

Switch case

switch(expression) {

case value1:

==>

break;

case value2:

==>

break;

default:

==>

}

day = "fri" ?
cases

```
switch (day) {  
  case "mon":  
    console.log("1st day of the week");  
    break;  
  case "tue":  
    console.log("2nd day of the week");  
    break;  
}
```

```
const day = "fri";
```

```
switch (day) {
```

```
  ✓ case "mon":  
    console.log("1st day of the week");
```

```
  ✓ case "tue":  
    console.log("2nd day of the week");
```

```
  ✓ case "wed":  
    console.log("3rd day of the week");
```

```
  ✓ case "thu":  
    console.log("4th day of the week");
```

```
  ✓ case "fri":  
    console.log("5th day of the week"); → print
```

```
  case "sat":  
    console.log("6th day of the week"); → print
```

```
  case "sun":  
    console.log("7th day of the week"); → print
```

```
  default:  
    console.log("Wrong day"); → print
```

```
}
```

→ break;